

How Stable Are Regression Conclusions Across Subpopulations?

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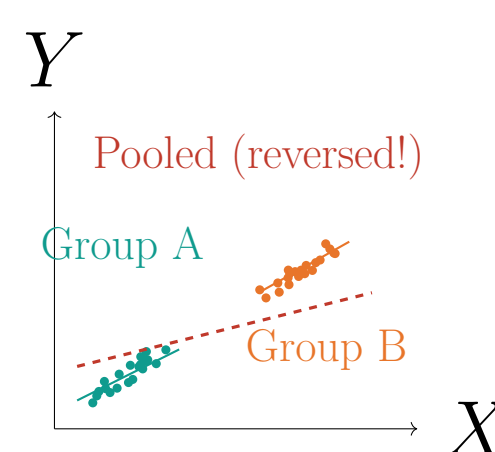
A FINITE-SAMPLE EMPIRICAL STUDY

The Problem in One Slide

You fit a regression on the full dataset. The coefficients look reasonable, R^2 is decent, p-values are small. A reviewer asks: *“Does this hold for men and women separately? For young and old? For each site?”*

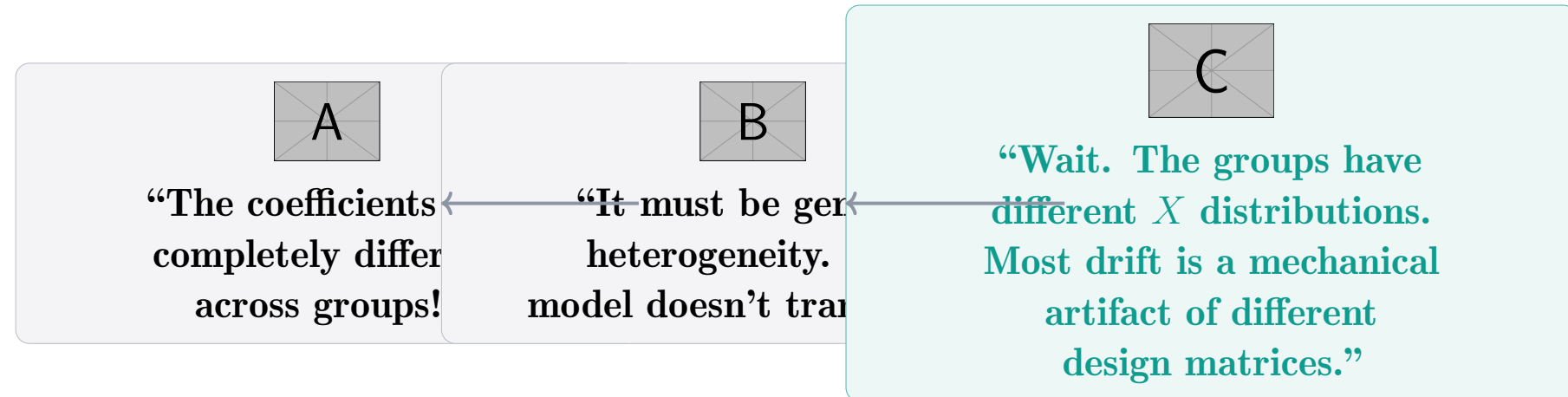
You check. It doesn’t. Now what?

Simpson’s Paradox: The Classic Warning



Within each group: Y increases with X (positive slope).
Pooled together: the trend *reverses*.
The pooled regression hides subgroup structure, and the sign of the coefficient depends on which groups you aggregate.
Our question: how often does this happen in practice, and can we detect it?

“It’s an Artifact” – The Core Intuition



The thesis: Observed coefficient differences have **two sources**: genuine parameter heterogeneity, and mechanical artifacts from covariate distribution differences. Standard practice conflates them. We propose a simple decomposition to separate the two.

Study Design: Four Models Compared

For each dataset or simulated replicate we fit four nested specifications:

Model	Specification
M1 (Pooled)	$Y = \beta_0 + X^\top \beta + \varepsilon$
M2 (Intercept)	$Y = \beta_0 + X^\top \beta + \gamma G + \varepsilon$
M3 (Interaction)	$Y = \beta_0 + X^\top \beta + \gamma G + X^\top \delta G + \varepsilon$
M4 (Separate)	$Y = \beta_{0,g} + X^\top \beta_g + \varepsilon$ (per group)

This progression isolates: (i) intercept shift, (ii) slope heterogeneity, (iii) full subgroup structure.

Five Stability Metrics

Metric	Definition	What it captures
Sign Stability Rate (SSR)	$SSR_j = \frac{1}{M} \sum_m \mathbf{1}(\text{sgn } \hat{\beta}_{j,a_m} = \text{sgn } \hat{\beta}_{j,b_m})$	Do substantive conclusions (+/–) reverse across subgroups?
Standardized Drift	$D_j^* = \frac{\max_g \hat{\beta}_{j,g} - \hat{\beta}_{j,\text{pooled}} }{SE(\hat{\beta}_{j,\text{pooled}})}$	How far do subgroup coefficients move from pooled, in SE units?
Context-Normalized Drift	$\tilde{D}_j = \frac{D_j^*}{1 + M_j}$ $M_j = \frac{1}{\binom{G}{2}} \sum_{a < b} \frac{ \bar{x}_{j,a} - \bar{x}_{j,b} }{s_{j,\text{pooled}}}$	Drift after accounting for covariate distribution differences. The key diagnostic.
Transported Prediction Gap	$TPG_{a \rightarrow b} = \text{RMSE}_{\text{pooled} \rightarrow b} - \text{RMSE}_{b \rightarrow b}$	How much prediction error increases when a model travels across groups.
Covariate Shift Summary	$CSS_{a,b} = \frac{1}{p} \sum_j \frac{ \bar{x}_{j,a} - \bar{x}_{j,b} }{s_{j,\text{pooled}}}$	How different are the X distributions between groups?

The Key Decomposition: Why Context-Normalized Drift Is Smaller

Under OLS, even with **identical** true β across groups:

$$\hat{\beta}^{(g)} - \hat{\beta}^{(\text{pooled})} = (X'_g X_g)^{-1} X'_g \varepsilon_g - (X'X)^{-1} X' \varepsilon$$

The variance: $\text{Var}(\hat{\beta}^{(g)} - \hat{\beta}^{(\text{pooled})}) = \sigma^2[(X'_g X_g)^{-1} + (X'X)^{-1}]$.
When group g has a different X distribution, $(X'_g X_g)^{-1} \neq (X'X)^{-1}$, and the difference inflates **even when β is identical**. This is the **mechanical** component of drift.

$$\underbrace{D_j^*}_{\text{Total observed drift}} = \underbrace{\tilde{D}_j}_{\text{Residual (unexplained by } X)} \times \underbrace{(1 + M_j)}_{\text{Mechanical inflation from covariate spread}}$$

Three diagnostic regimes:

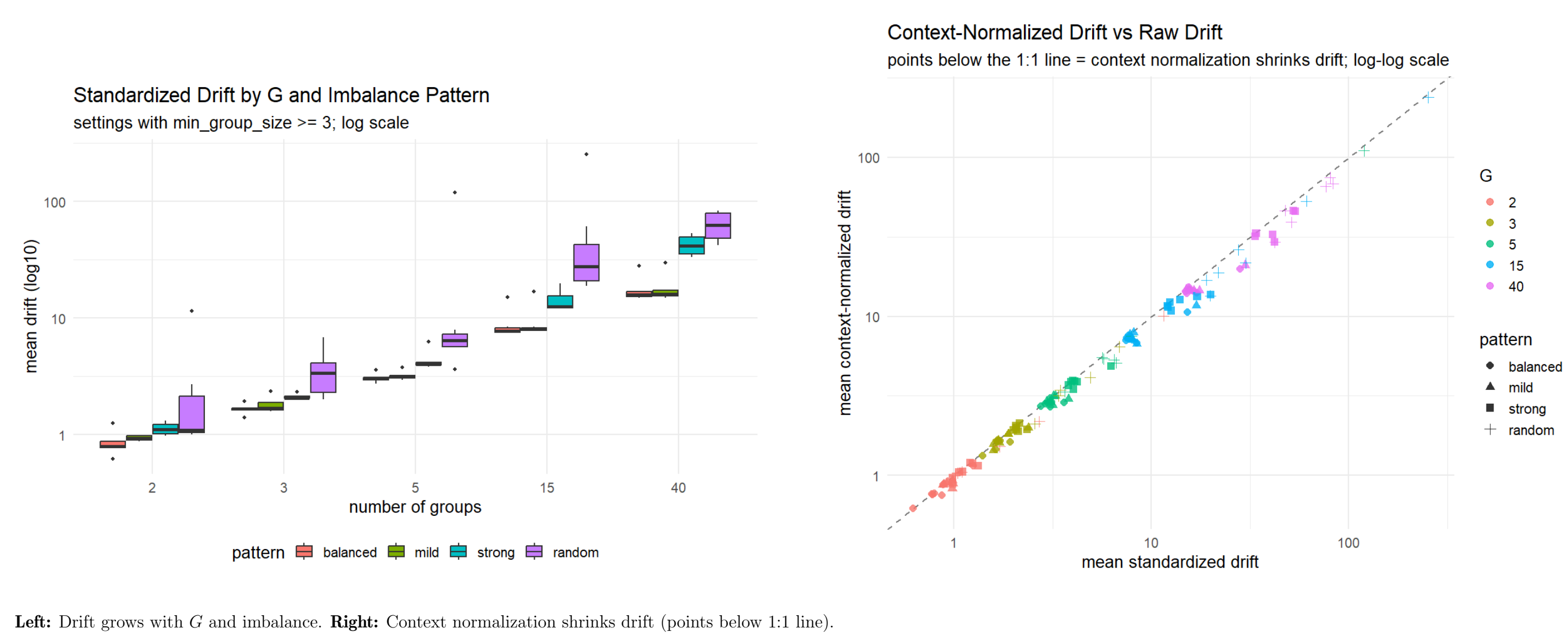
$M_j \approx 0$: $\tilde{D}_j \approx D_j^*$. No correction needed. All drift is “real.”

M_j large, $\tilde{D}_j$ small: Drift is an **artifact** of different X distributions. Reassuring.

M_j large, $\tilde{D}_j$ still large: Genuine parameter heterogeneity, confirmed after correction.

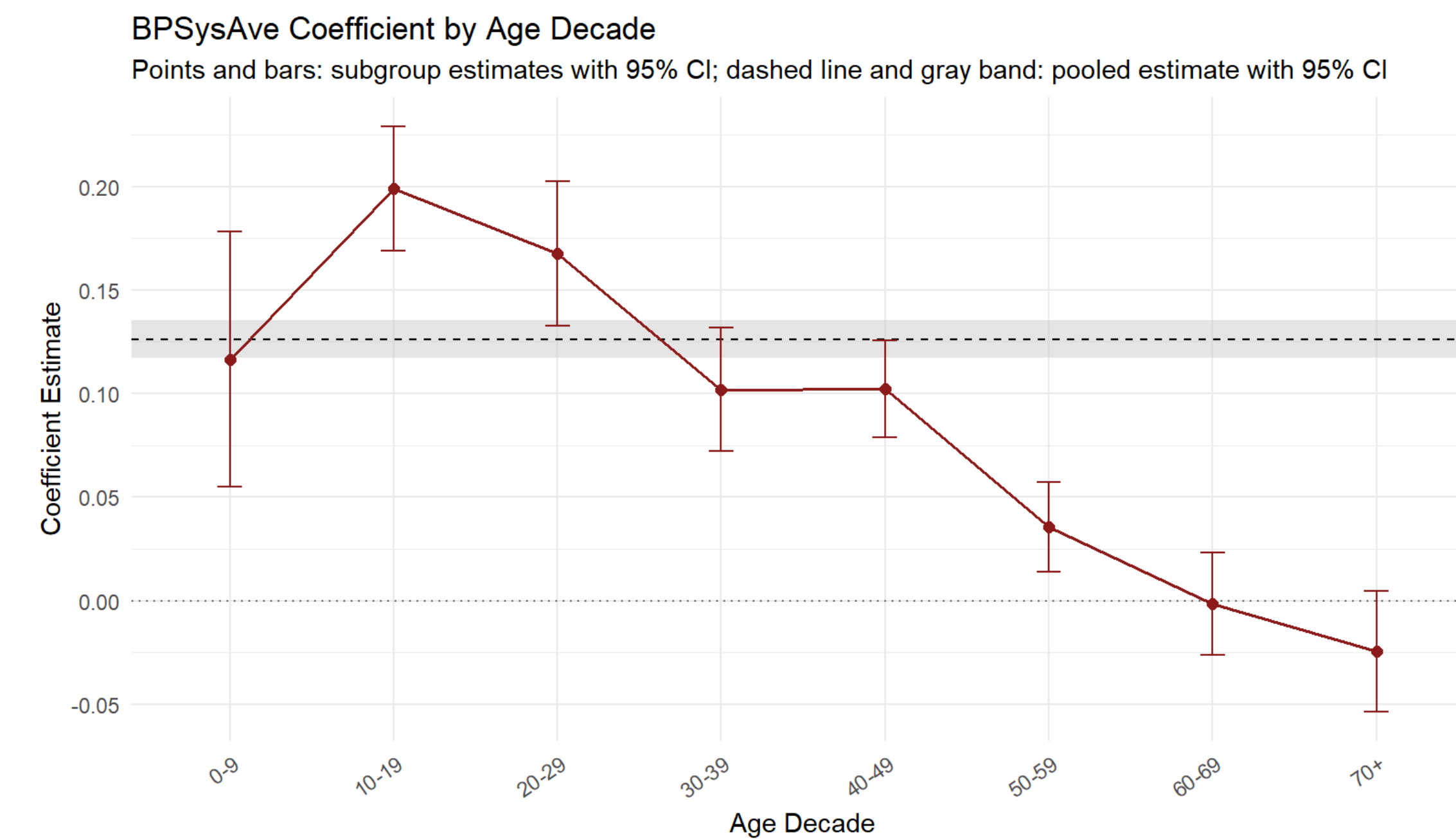
Simulation Grid: 140 Settings (Scenario 1)

Factors varied: $G \in \{3, 5, 10, 20, 40\}$ groups; total $n \in \{100, 300, 1000, 3000, 10000, 30000\}$; 4 imbalance patterns (balanced, mild, strong, Dirichlet-random).
Each setting: 200 Monte Carlo replicates, 4 models fitted, 5 metrics computed with means, SDs, and 95% CIs.



Real-Data Evidence: NHANES Blood Pressure

NHANES 2017–2018: $n = 9,254$, response = systolic BP (BPSysAve), subgroups = age decades (20–29 through 70+), predictors = BMI, waist circumference, total cholesterol, HDL.

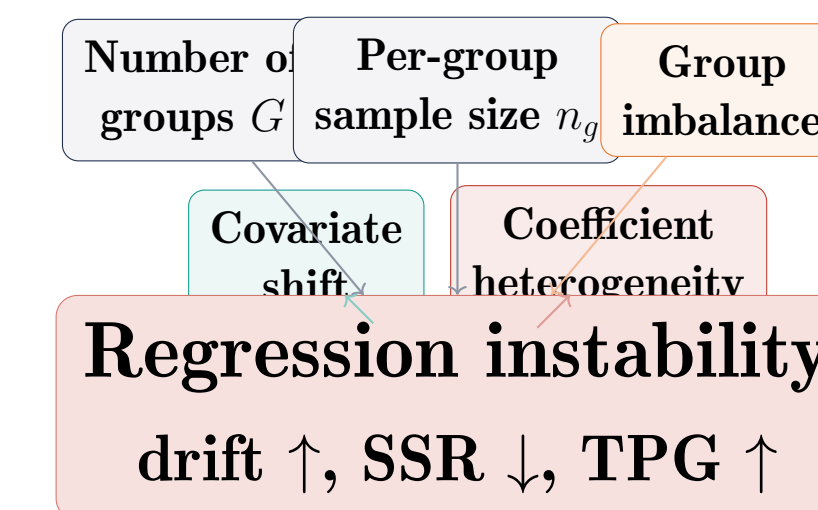


Coefficient CIs by age decade; note the reversal in the 60–69 and 70+ groups for waist circumference.

Metric	Value
SSR (all predictors)	0.57
Max standardized drift	32.07 (BPSysAve)
Context-norm. drift	7.49 (BPSysAve)
Avg. CSS (pairwise)	0.28

Key finding: Waist circumference’s effect on blood pressure *reverses sign* in older age groups. Standardized drift of 32 is extreme, but context normalization brings it to 7.5 – still large, confirming genuine heterogeneity, not just covariate shift.

What Drives Regression Instability?



Simulation findings (Scenario 1, no heterogeneity):

- Drift is driven by G and per-group n , not total N .
- Strong/random imbalance amplifies drift 2–4× over balanced.
- SSR stays at 1.0 – with identical β , there is no sign reversal.
- Context normalization systematically reduces drift; shrinkage $\propto$ CSS.

Take-Home Messages

- Regression conclusions are fragile.** Subgroup-specific estimates routinely disagree with pooled estimates in direction, magnitude, and significance.
- Not all drift is real.** Covariate distribution differences across groups mechanically inflate coefficient differences. Context-normalized drift separates artifact from genuine instability.
- Simpson’s paradox is not a curiosity.** It is a special case of a general